MATH1025B
Essential Mathematical Methods

Chapter 2: Mathematical Induction

September 11, 2025

Statement for each positive integer

Consider the expressions $1^3 + 2^3 + \cdots + n^3$ and $\left(\frac{n \cdot (n+1)}{2}\right)^2$.

$n = 1$	1^3	$= 1$	$\left(\frac{1 \cdot 2}{2}\right)^2 = 1$
$n = 2$	$1^3 + 2^3$	$= 9$	$\left(\frac{2 \cdot 3}{2}\right)^2 = 9$
$n = 3$	$1^3 + 2^3 + 3^3$	$= 36$	$\left(\frac{3 \cdot 4}{2}\right)^2 = 36$
$n = 4$	$1^3 + 2^3 + 3^3 + 4^3$	$= 100$	$\left(\frac{4 \cdot 5}{2}\right)^2 = 100$
$\vdots$	$\vdots$		$\vdots$

It seems that $1^3 + 2^3 + \cdots + n^3 = \left(\frac{n \cdot (n+1)}{2}\right)^2$ for all positive integers n.

Question

Is it really true in general? If so, how can we justify it?

Mathematical Induction

A **statement** is a declarative sentence or meaningful string of symbols that can be classified as being true or false.

Theorem 2.1 (Principle of Mathematical Induction)

Let $P(n)$ be a statement for each $n \in \mathbb{Z}^+$. Suppose

(i) *$P(1)$ is true; and*

(ii) *for any integer $k \geq 1$, if $P(k)$ is true then $P(k+1)$ is true.*

Then $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Idea:

- From (i), $P(1)$ is true.
- From (ii), $P(1)$ is true implies that $P(2)$ is true as well.
- From (ii) again, $P(2)$ is true implies that $P(3)$ is true as well. And so on.

Format of argument by mathematical induction

To argue the statement "for any $n \in \mathbb{Z}^+$, $P(n)$ is true", we may proceed as follows:

- **Step 0.** Identify $P(n)$ and write it down explicitly.
- **Step 1.** (The **base case**): Verify the statement $P(1)$.
- **Step 2.** (The **induction step**): Declare that we assume $P(k)$ to be true for a particular k. (This is called the **induction assumption**.) Based on such an assumption, deduce that $P(k+1)$ is true.
- **Step 3.** Declare that according to the Principle of Mathematical Induction, $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Example 2.1

Apply mathematical induction to verify the statement

$$1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4} \quad \text{for any } n \in \mathbb{Z}^+.$$

- For each $n \in \mathbb{Z}^+$, denote by $P(n)$ the statement

$$\text{“}1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}\text{”}.$$

- When $n = 1$,

$$\text{L.H.S.} = 1^3 = \frac{1^2(1+1)^2}{4} = \text{R.H.S.}$$

  So $P(1)$ is true.

Example 2.1

Apply mathematical induction to verify the statement

$$1^3 + 2^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4} \quad \text{for any } n \in \mathbb{Z}^+.$$

- Assume $P(k)$ is true where $k \geq 1$, i.e. $1^3 + 2^3 + \cdots + k^3 = \dfrac{k^2(k+1)^2}{4}$. Then,

$$\begin{aligned} 1^3 + 2^3 + \cdots + k^3 + (k+1)^3 &= \frac{k^2(k+1)^2}{4} + (k+1)^3 \\ &= \frac{(k+1)^2(k^2 + 4(k+1))}{4} \\ &= \frac{(k+1)^2(k+2)^2}{4}. \end{aligned}$$

  Hence $P(k+1)$ is true.
- By the Principle of Mathematical Induction, $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Recall: Let $m, n \in \mathbb{Z}$. m is said to be **divisible** by n, denoted by $n \mid m$, if there exists $k \in \mathbb{Z}$ such that $m = kn$.

Example 2.2

Prove that $n^3 - n$ is divisible by 3 for any $n \in \mathbb{Z}^+$.

- For each $n \in \mathbb{Z}^+$, denote by $P(n)$ the statement "$3 \mid n^3 - n$".
- When $n = 1$,
$$1^3 - 1 = 0 = 3 \times 0.$$
So $P(1)$ is true.
- Assume $P(k)$ is true where $k \geq 1$, i.e. $3 \mid k^3 - k$, which is equivalent to $k^3 - k = 3m$ for some $m \in \mathbb{Z}$. Then,
$$\begin{aligned}(k+1)^3 - (k+1) &= (k^3 + 3k^2 + 3k + 1) - (k+1) \\ &= (k^3 - k) + 3k^2 + 3k \\ &= 3(m + k^2 + k).\end{aligned}$$
Since $m, k \in \mathbb{Z}$, we have $m + k^2 + k \in \mathbb{Z}$ and thus $3 \mid (k+1)^3 - (k+1)$. Hence $P(k+1)$ is true.
- By the Principle of Mathematical Induction, $P(n)$ is true for all $n \in \mathbb{Z}^+$.

"False proof" by induction

The following argument "appears" to prove that all horses are of the same color by mathematical induction.

- For each $n \in \mathbb{Z}^+$, let $P(n)$ denote the statement "In every set of horses of size n, all n horses are the same color".
- $P(1)$ is clearly true.
- Assume $P(n)$ is true where $n \geq 1$.
  Let $\{h_1, h_2, \ldots, h_n, h_{n+1}\}$ be a set of $n+1$ horses.
  Then each of the sets $\{h_1, h_2, \ldots, h_n\}$, $\{h_2, \ldots, h_n, h_{n+1}\}$ consists of n horses.
  By the induction assumption, there is only one color within each set.
  Since the two sets overlap, all $n+1$ horses are the same color, i.e. $P(n+1)$ is true.

Question

What is wrong with this argument?

By saying the two sets overlap, we implicitly assume that $n \geq 2$. In fact, the inductive step is false when $n = 1$.

Example 2.3

Prove the identity $\displaystyle\sum_{r=1}^{n} \frac{1}{\sin rx \sin(r+1)x} = \frac{\sin nx}{\sin^2 x \sin(n+1)x}$ for all $n \in \mathbb{Z}^+$.

- For $n \in \mathbb{Z}^+$, let $P(n)$ denote the statement

$$\text{“}\sum_{r=1}^{n} \frac{1}{\sin rx \sin(r+1)x} = \frac{\sin nx}{\sin^2 x \sin(n+1)x}\text{”}.$$

- When $n = 1$,

$$\text{L.H.S.} = \frac{1}{\sin x \sin 2x} \quad \text{and}$$
$$\text{R.H.S.} = \frac{\sin x}{\sin^2 x \sin 2x} = \frac{1}{\sin x \sin 2x}.$$

  So $P(1)$ is true.

- Assume $P(k)$ is true where $k \geq 1$, i.e.

$$\sum_{r=1}^{k} \frac{1}{\sin rx \sin(r+1)x} = \frac{\sin kx}{\sin^2 x \sin(k+1)x}.$$

Example 2.3

Prove the identity $\displaystyle\sum_{r=1}^{n} \frac{1}{\sin rx \sin(r+1)x} = \frac{\sin nx}{\sin^2 x \sin(n+1)x}$ for all $n \in \mathbb{Z}^+$.

Then,

$$\begin{aligned}
\sum_{r=1}^{k+1} \frac{1}{\sin rx \sin(r+1)x} &= \sum_{r=1}^{k} \frac{1}{\sin rx \sin(r+1)x} + \frac{1}{\sin(k+1)x \sin(k+2)x} \\
&= \frac{\sin kx}{\sin^2 x \sin(k+1)x} + \frac{1}{\sin(k+1)x \sin(k+2)x} \\
&= \frac{\sin kx \sin(k+2)x + \sin^2 x}{\sin^2 x \sin(k+1)x \sin(k+2)x} \\
&= \frac{-\frac{1}{2}\left[\cos(2k+2)x - \cos(2x)\right] + \sin^2 x}{\sin^2 x \sin(k+1)x \sin(k+2)x} \\
&= \frac{-\frac{1}{2}\left[1 - 2\sin^2(k+1)x - 1 + 2\sin^2 x\right] + \sin^2 x}{\sin^2 x \sin(k+1)x \sin(k+2)x} \\
&= \frac{\sin(k+1)x}{\sin^2 x \sin(k+2)x}.
\end{aligned}$$

Hence $P(k+1)$ is true.

- By the Principle of Mathematical Induction, $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Exercise 2.1

By Mathematical Induction, show that $P(n)$ is true for all $n \in \mathbb{Z}^+$ if

(a) $P(n) = \text{“}\sum_{r=1}^{n} r(r+1)(r+2) = \frac{1}{4}n(n+1)(n+2)(n+3)\text{”}$;

(b) $P(n) = \text{“}\sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)} = \frac{n(n+1)}{2(2n+1)}\text{”}$;

(c) $P(n) = \text{“}5 \mid 3^{2n} - 2^{2n}\text{”}$;

(d) $P(n) = \text{“}64 \mid 9^n - 8n - 1\text{”}$;

(e) $P(n) = \text{“}\prod_{r=0}^{n} \cos(2^r\theta) = \frac{\sin 2^{n+1}\theta}{2^{n+1}\sin\theta}\text{”}$ (revised);

(f) $P(n) = \text{“}\sum_{r=1}^{n} \sin rx = \frac{\sin\left(\frac{n+1}{2}\right)x \sin\frac{nx}{2}}{\sin\frac{x}{2}}\text{”}$.

Exercise 2.2

For any positive integer n, a $2^n \times 2^n$ checkerboard with any one square removed can be tiled using right triominoes.

Variations of Principle of Mathematical Induction

Theorem 2.2 (Base case other than $n = 1$)

Let $n_0 \in \mathbb{Z}$ and let $P(n)$ be a statement for each integer $n \geq n_0$. Suppose

(i) *$P(n_0)$ is true; and*

(ii) *for any integer $k \geq n_0$, if $P(k)$ is true then $P(k+1)$ is true.*

Then $P(n)$ is true for all integers $n \geq n_0$.

Idea:

- From (i), $P(n_0)$ is true.
- From (ii), $P(n_0)$ is true implies that $P(n_0 + 1)$ is true as well.
- From (ii) again, $P(n_0 + 1)$ is true implies that $P(n_0 + 2)$ is true as well. And so on.

Example 2.4 (n^2 versus 2^n)

Show that for any integer $n \geq 5$, $n^2 < 2^n$.

- For $n \in \mathbb{Z}^+$, let $P(n)$ be the statement "$n^2 < 2^n$".
- When $n = 5$, $5^2 = 25 < 32 = 2^5$. So $P(5)$ is true.
- Assume that $P(k)$ is true where $k \geq 5$. Then,

$$\begin{aligned}
(k+1)^2 &= k^2 + 2k + 1 \\
&< k^2 + 2k + k && (\because 1 < 5 \leq k) \\
&= k^2 + 3k \\
&< k^2 + k^2 && (\because 3k < 5k \leq k^2) \\
&= 2k^2 \\
&< 2 \cdot 2^k && \text{(by the inductive hypothesis)} \\
&= 2^{k+1}.
\end{aligned}$$

  Hence $P(k+1)$ is true.
- By the Principle of Mathematical Induction, $P(n)$ is true for all integers $n \geq 5$.

Variations of Principle of Mathematical Induction

Theorem 2.3

Let $P(n)$ be a statement for each $n \in \mathbb{Z}^+$. Suppose

(i) *$P(1)$ is true (respectively $P(2)$ is true); and*

(ii) *for any integer $k \geq 1$ (respectively $k \geq 2$), if $P(k)$ are true then $P(k+2)$ is true.*

Then $P(n)$ is true for all positive odd (respectively even) numbers n.

Exercise 2.3

(a) Prove that $23 \mid 12^n - 11^n$ for all positive even numbers n.

(b) Prove that $11 \mid 7^n + 4^n$ for all positive odd numbers n.

Variations of Principle of Mathematical Induction

Theorem 2.4

Let $P(n)$ be a statement for each $n \in \mathbb{Z}^+$. Suppose

(i) *$P(1)$ and $P(2)$ are true; and*

(ii) *for any integer $k \geq 1$, if $P(k)$ and $P(k+1)$ are true then $P(k+2)$ is true.*

Then $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Idea:

- From (i), $P(1), P(2)$ are true.
- From (ii), $P(1), P(2)$ are true implies that $P(3)$ is true.
- From (ii) again, $P(2), P(3)$ are true implies that $P(4)$ is true as well. And so on.

Example 2.5

Prove that $2^n \mid (3+\sqrt{5})^n + (3-\sqrt{5})^n$ for all $n \in \mathbb{Z}^+$.

Set $\alpha = 3+\sqrt{5}$ and $\beta = 3-\sqrt{5}$. Then $\alpha+\beta = 6$, $\alpha\beta = 4$ and

$$\alpha^2 - 6\alpha + 4 = 0, \qquad \beta^2 - 6\beta + 4 = 0.$$

- For each $n \in \mathbb{Z}^+$, let $P(n)$ be the statement "$2^n \mid \alpha^n + \beta^n$".
- Since $\alpha+\beta = 6 = 2\times 3$ and $\alpha^2+\beta^2 = 28 = 2^2\times 7$, so $P(1)$ and $P(2)$ are true.
- Assume $P(k)$ and $P(k+1)$ are true where $k \geq 1$. That is, $\alpha^k + \beta^k = 2^k M$ and $\alpha^{k+1} + \beta^{k+1} = 2^{k+1}N$ for some $M, N \in \mathbb{Z}$. Then,

$$\begin{aligned}\alpha^{k+2} + \beta^{k+2} &= (\alpha^{k+1}+\beta^{k+1})(\alpha+\beta) - \alpha\beta(\alpha^k+\beta^k)\\ &= 2^{k+1}N\cdot 6 - 4\cdot 2^k M\\ &= 2^{k+2}(3N-M), \qquad \text{where } 3N-M \in \mathbb{Z}.\end{aligned}$$

  Hence $P(k+2)$ is true.
- By the Principle of Mathematical Induction, $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Example 2.6

Prove that if only two types of coins, a 3-cent coin and a 5-cent coin, are minted, then any amount of money greater than 7 cents can be paid in coins.

- For any integer $k > 7$, denote by $P(k)$ the statement below

  there exist some integers $n, m \geq 0$ such that $k = 3n + 5m$.

- Note that

$$\begin{aligned} 8 &= 3 + 5 \\ 9 &= 3 + 3 + 3 \\ 10 &= 5 + 5. \end{aligned}$$

  So $P(8), P(9), P(10)$ are true.
- Assume that $P(k), P(k+1), P(k+2)$ are true where $k \geq 8$. Then there exist some integers $n, m \geq 0$ such that $k = 3n + 5m$. And thus $k + 3 = 3(n+1) + 5m$. So $P(k+3)$ is true.
- By the Principle of Mathematical Induction, $P(k)$ is true for all integers $k > 7$.

Variations of Principle of Mathematical Induction

Theorem 2.5 (Strong Mathematical Induction)

Let $n_0 \in \mathbb{Z}$ and let $P(n)$ be a statement for each integer $n \geq n_0$. Suppose

(i) *$P(n_0)$ is true; and*

(ii) *for any integer $k \geq n_0$, if $P(n_0), P(n_0+1), \ldots, P(k)$ are true then $P(k+1)$ is true.*

Then $P(n)$ is true for all integers $n \geq n_0$.

Idea:

- From (i), $P(n_0)$ is true.
- From (ii), $P(n_0)$ is true implies that $P(n_0+1)$ is true.
- From (ii) again, $P(n_0), P(n_0+1)$ are true implies that $P(n_0+2)$ is true.
- From (ii) again, $P(n_0), P(n_0+1), P(n_0+2)$ are true implies that $P(n_0+3)$ is true. And so on.

Example 2.7 (Prime factorization)

Show that every positive integer $n \geq 2$ can be expressed as a product of one or more prime numbers.

- Let $P(n)$ be the statement "n can be expressed as a product of one or more prime numbers".
- $P(2)$ is trivially true since 2 is a prime itself.
- Assume that $P(2), P(3), \ldots, P(k)$ are true where $k \geq 2$, i.e. every integer r with $2 \leq r \leq k$ can be expressed as a product of primes.
  - Case 1: $k+1$ is a prime. Then it is a product of one prime, i.e. itself.
  - Case 2: $k+1$ is not a prime. Then it must be composite, i.e.
    $$k+1 = ab \text{ for some integers } a, b \text{ such that } 2 \leq a, b \leq k.$$
    By the induction assumption, a can be written as a product of primes $p_1 p_2 \cdots p_i$ and b can be written as a product of primes $q_1 q_2 \cdots q_j$.
    So $k+1$ can be written as a product of primes $p_1 p_2 \cdots p_i q_1 q_2 \cdots q_j$.

  So $P(k+1)$ is true.
- By the Principle of Mathematical Induction, $P(n)$ is true for all positive integers $n \geq 2$.

Example 2.8

Prove that every positive integer can be written as a sum of distinct nonnegative integer powers of 2.

- For $n \in \mathbb{Z}^+$, let $P(n)$ be the statement "there exist distinct nonnegative integers $r_1, r_2, \ldots, r_p$ such that
$$n = 2^{r_1} + 2^{r_2} + \cdots + 2^{r_p}\text{''}.$$

- $P(1)$ is clearly true since $1 = 2^0$.

- Assume $P(1), \ldots, P(k)$ are true where $k \geq 1$.
Let ℓ be the largest nonnegative integer such that $2^\ell \leq k+1$.
Let $m = k + 1 - 2^\ell$. Then m is a nonnegative integer satisfying $m < k+1$ (since $2^\ell \geq 1$).
By the strong inductive hypothesis, there are distinct integers $r_1, r_2, \ldots, r_p \geq 0$ such that $m = 2^{r_1} + 2^{r_2} + \cdots + 2^{r_p}$.
Thus we have
$$k + 1 = 2^{r_1} + 2^{r_2} + \cdots + 2^{r_p} + 2^\ell.$$

Example 2.8

Prove that every positive integer can be written as a sum of distinct nonnegative integer powers of 2.

Having proved that $k+1 = 2^{r_1} + 2^{r_2} + \cdots + 2^{r_p} + 2^{\ell}$, it remains to show that $\ell \neq r_j$ for all $j = 1, 2, \ldots, p$.
Suppose, for the sake of contradiction, that $\ell = r_j$ for some j.
Then $m \geq 2^{r_j} = 2^{\ell}$ and hence $k+1 = m + 2^{\ell} \geq 2^{\ell} + 2^{\ell} = 2^{\ell+1}$.
This contradicts the fact that ℓ is the largest nonnegative integer such that $2^{\ell} \leq k+1$.
Hence $r_1, r_2, \ldots, r_p, \ell$ are all distinct and $P(k+1)$ is true.

- By the Principle of Mathematical Induction, $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Example 2.9

Breaking a rectangular chocolate bar with $n \times m$ pieces into individual pieces requires $nm - 1$ breaks.

Let $P(n, m)$ be the number of breaks needed to split up $n \times m$ squares. We argue by induction on the sum $n + m$.

Base Case: For a 1×1 bar, we are already done, so no breaks are needed. $P(1, 1) = 0 = 1 \times 1 - 1$.

Induction Step: Suppose $P(i, j) = ij - 1$ for any $i, j \in \mathbb{Z}^+$ with $i + j < n + m$. Given an $n \times m$ bar, we may assume that the first break is along a row, and we get an $n_1 \times m$ bar and an $n_2 \times m$ bar, where $n_1 + n_2 = n$. By the induction assumption, the numbers of further breaks we need for each of the bars are $n_1 m - 1$ and $n_2 m - 1$. Hence, the total number of breaks is

$$P(n, m) = 1 + (n_1 m - 1) + (n_2 m - 1) = (n_1 + n_2)m - 1 = nm - 1.$$

Exercise 2.4

Suppose a particular real number x has the property that $x + \frac{1}{x}$ is an integer. Prove that $x^n + \frac{1}{x^n}$ is an integer for all natural numbers n.

Exercise 2.5

Every amount of postage that is at least 12 cents can be made from 4-cent and 5-cent stamps.

Exercise 2.6

Prove, using strong induction, that every natural number is either a Fibonacci number or can be written as the sum of distinct Fibonacci numbers.

Exercise 2.7

The Fibonacci sequence $F_0, F_1, F_2, F_3, \ldots$ is a sequence of integers defined by the recurrence relation

$$F_0 = 0, \quad F_1 = 1 \quad \text{and} \quad F_{n+2} = F_{n+1} + F_n \quad \text{for } n \geq 0.$$

(a) Prove that $\sum_{i=0}^{n} F_i^2 = F_n F_{n+1}$ for all $n \geq 0$.

(b) Prove that the Fibonacci sequence satisfies the identity

$$F_n F_m + F_{n+1} F_{m+1} = F_{n+m+1} \qquad \text{for all integers } n, m \geq 0.$$

(c) Let φ and ψ ($\varphi > \psi$) be the roots of the equation $x^2 - x - 1 = 0$. Prove that $F_n = \frac{1}{\sqrt{5}} (\varphi^n - \psi^n)$ for all $n \geq 0$.

Theorem 2.6 (Forward-Backward Induction)

Let $P(n)$ be a statement for each $n \in \mathbb{Z}^+$. Suppose

(i) *$P(1)$ is true; and*

(ii) *for any $k \geq 1$, if $P(k)$ is true then $P(2k)$ is true; and*

(iii) *for any $k \geq 2$, if $P(k)$ is true then $P(k-1)$ is true.*

Then $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Example 2.10

Let $n \in \mathbb{Z}^+$ and let $x_1, x_2, \ldots, x_n$ be n positive real numbers. Prove that

$$\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right)^2 \leq \frac{x_1^2 + x_2^2 + \cdots + x_n^2}{n}.$$

When $n = 2$:

$$(x_1 - x_2)^2 \geq 0 \implies x_1^2 - 2x_1x_2 + x_2^2 \geq 0 \implies 2(x_1^2 + x_2^2) \geq x_1^2 + 2x_1x_2 + x_2^2.$$

Hence

$$2(x_1^2 + x_2^2) \geq (x_1 + x_2)^2.$$

Variations of Principle of Mathematical Induction

- For each $n \in \mathbb{Z}^+$, let $P(n)$ denote the statement "Given n positive real numbers $x_1, x_2, \ldots, x_n$,

$$\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right)^2 \leq \frac{x_1^2 + x_2^2 + \cdots + x_n^2}{n} \text{ ".}$$

- $P(1)$ is trivially true since L.H.S. $= x_1^2 =$ R.H.S..
- Assume $P(k)$ is true where $k \geq 1$. Then,

$$\begin{aligned}
&\left(\frac{x_1 + x_2 + \cdots + x_{2k}}{2k}\right)^2 \\
&= \frac{1}{4}\left(\frac{x_1 + \cdots + x_k}{k} + \frac{x_{k+1} + \cdots + x_{2k}}{k}\right)^2 \\
&\leq \frac{1}{2}\left(\frac{x_1 + \cdots + x_k}{k}\right)^2 + \frac{1}{2}\left(\frac{x_{k+1} + \cdots + x_{2k}}{k}\right)^2 && (\because (x+y)^2 \leq 2(x^2+y^2)) \\
&\leq \frac{1}{2}\left(\frac{x_1^2 + \cdots + x_k^2}{k}\right) + \frac{1}{2}\left(\frac{x_{k+1}^2 + \cdots + x_{2k}^2}{k}\right) && \text{(by induction hypothesis)} \\
&= \frac{x_1^2 + \cdots + x_k^2 + x_{k+1}^2 + \cdots + x_{2k}^2}{2k}.
\end{aligned}$$

Hence $P(2k)$ is true.

Variations of Principle of Mathematical Induction

- Assume $P(k)$ is true where $k \geq 2$, i.e. given $y_1, y_2, \ldots, y_k > 0$, $\left(\frac{y_1+y_2+\cdots+y_k}{k}\right)^2 \leq \frac{y_1^2+y_2^2+\cdots+y_k^2}{k}$.
  Suppose $x_1, x_2, \ldots, x_{k-1}$ are $k-1$ positive real numbers. Put

  $$y_1 = x_1, \quad y_2 = x_2, \quad \ldots \quad y_{k-1} = x_{k-1}, \text{ and } y_k = \frac{x_1 + x_2 + \cdots + x_{k-1}}{k-1}.$$

  Then

  $$y_1 + y_2 + \cdots + y_k = \frac{k}{k-1}(x_1 + x_2 + \cdots + x_{k-1}),$$

  and so, by induction assumption,

  $$\begin{aligned}\left(\frac{x_1 + x_2 + \cdots + x_{k-1}}{k-1}\right)^2 &= \left(\frac{y_1 + y_2 + \cdots + y_k}{k}\right)^2 \leq \frac{y_1^2 + y_2^2 + \cdots + y_k^2}{k} \\ &= \frac{x_1^2 + x_2^2 + \cdots + x_{k-1}^2 + \left(\frac{x_1+x_2+\cdots+x_{k-1}}{k-1}\right)^2}{k}.\end{aligned}$$

  Therefore

  $$\left(\frac{x_1 + x_2 + \cdots + x_{k-1}}{k-1}\right)^2 \leq \frac{x_1^2 + x_2^2 + \cdots + x_{k-1}^2}{k-1}.$$

  Hence $P(k-1)$ is true.
- By Forward-Backward Induction, $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Exercise 2.8

Let $n \in \mathbb{Z}^+$ and let $x_1, x_2, \ldots, x_n$ be n positive real numbers. Suppose $m \in \mathbb{Z}^+$. Prove that

$$\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right)^m \leq \frac{x_1^m + x_2^m + \cdots + x_n^m}{n}.$$

Hint: Can you show that

$$\left(\frac{x+y}{2}\right)^m \leq \frac{x^m + y^m}{2} \qquad \text{whenever } x, y \geq 0,\ m \in \mathbb{Z}^+.$$